

PROJECT EXECUTABLE FILES

Date	15 March 2026
Team ID	SWTID-2026-8809
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Team Members	N.Pooja, T.Sangeetha, K.Divya, Veronica, N.Monika
Maximum Marks	3 Marks

Project Overview & Description

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI. Designed for students of all academic levels, EduGenie enables users to ask questions, receive simplified explanations of complex concepts, generate custom quizzes, obtain structured learning paths, and summarize large passages. Built with **FastAPI** and an **HTML/CSS** interface, it leverages **Google Gemini 1.5 Pro** for cloud intelligence and **LaMini-Flan-T5** for local lightweight execution.

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA

S.No	Item to Submit	Submitted (Yes / No / NA)
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
EduGenie/
|
├── main.py                # FastAPI app & REST endpoints (/qa, /explain, /quiz, /summarize, /
learn/recommendations)
├── explanation_module.py  # Concept explanation logic using LaMini-Flan-T5-783M
├── qna.py                 # Question answering module using Gemini 1.5 Pro
├── quiz_module.py         # Quiz generation module using Gemini 1.5 Pro
├── summary_module.py     # Text summarization module using Gemini 1.5 Pro
├── learning_path.py      # Personalized learning recommendations generator
├── templates/
|   └── index.html        # Web UI frontend interface
├── static/
|   └── style.css         # Custom CSS styling
└── requirements.txt      # Python dependencies (fastapi, uvicorn, transformers, google-
generativeai, jinja2)
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://127.0.0.1:8000 (Local Uvicorn Server Deployment)
Login Credentials (Demo)	No authentication required for direct evaluator testing.
Platform / Hosting Provider	Local Uvicorn ASGI Server / Compatible with Render / AWS / GCP
Repository Link	https://github.com/Ravi-teja-777/AI-ML-and-GEN-AI-Track-Project-Template
Demo Video Link	Available upon request / Walkthrough included in final presentation

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

- Python 3.10+ installed and added to System PATH
- Google Gemini API Key (Obtained from Google AI Studio)
- Modern Web Browser (Chrome, Firefox, Safari, Edge)

Step 1: Clone or extract the repository

```
git clone <repository-url> && cd EduGenie
```

Step 2: Install required Python dependencies

```
pip install -r requirements.txt
```

Step 3: Configure environment variables

Set GEMINI_API_KEY in your environment or code configuration

Step 4: Start the application server

```
uvicorn main:app --reload
```

Step 5: Access the application

Open web browser and navigate to http://127.0.0.1:8000

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	First execution of local model (LaMini-Flan-T5) takes time to download weights.	Pre-download model weights or ensure stable internet during initial startup.

S.No	Known Issue / Limitation	Workaround / Status
2	Gemini API rate limits under heavy continuous request load.	Implemented error exception handling to notify users and safely return error state.
3	JSON parsing error on non-standard LLM outputs in Quiz generation.	Added regex-based markdown block cleaner (<code>clean_json_block`</code>) for reliable parsing.

Submitted By: Tella Divya Sree & Team

Mentor: Siri

Date of Submission: 11/04/2025

Status: Completed & Verified